

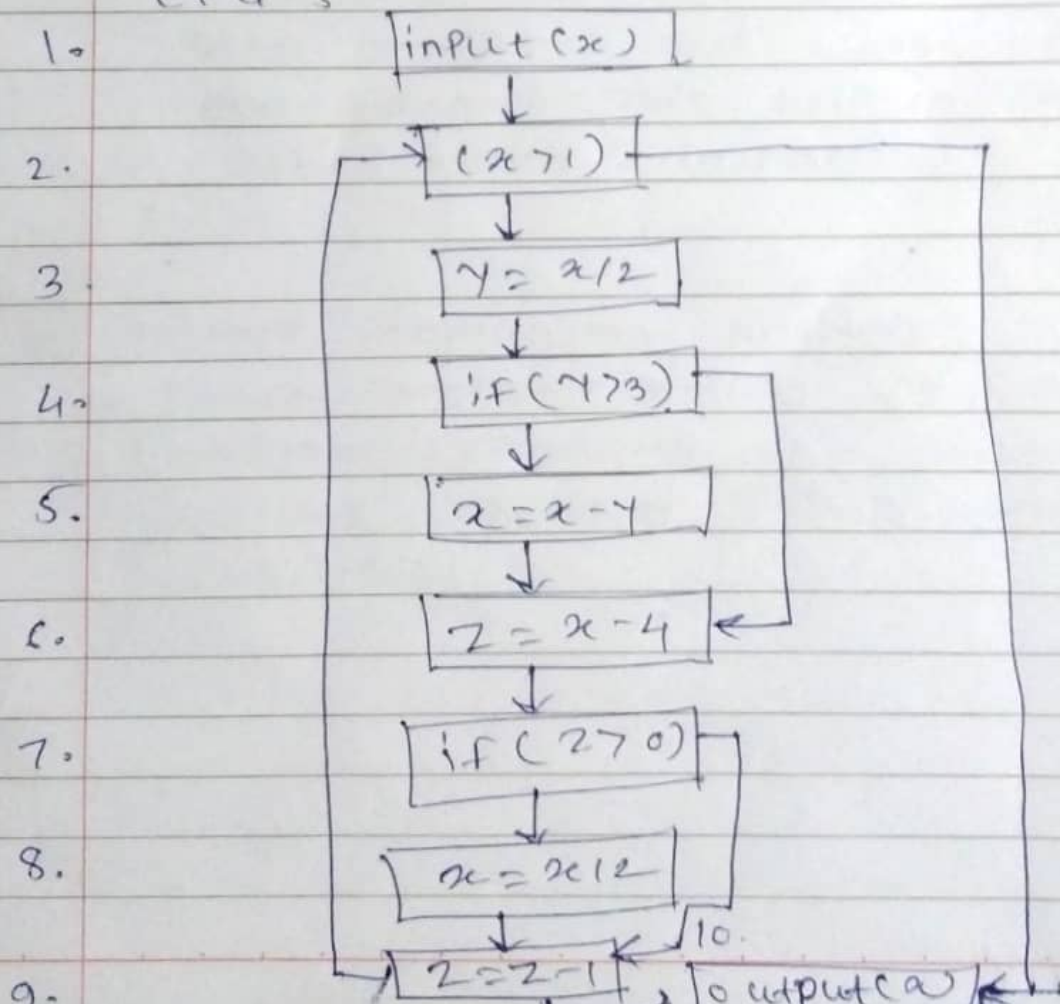
Q 1] Consider the following code. draw CFG and apply live variable analysis. Explain why constraint rules defined for assignment are sound?

```

→ input (x);
while (x > 1) {
    y = x/2;
    if (y > 3)
        x = x - y;
    z = x - 4;
    if (z > 0)
        x = x/2;
    z = z - 1;
    output (x);
}

```

CFG :-



Gen () and Kill () Sets :

statements	Gen()	Kill()
1.	$\emptyset$	$\{x\}$
2.	$\{x\}$	$\{\emptyset\}$
3.	$\{x\}$	$\{y\}$
4.	$\{y\}$	$\emptyset$
5.	$\{x, y\}$	$\{x\}$
6.	$\{x\}$	$\{z\}$
7.	$\{z\}$	$\emptyset$
8.	$\{x\}$	$\{x\}$
9.	$\{z\}$	$\{z\}$
10.	$\{x\}$	$\emptyset$

Entry and exit

transfer function :

$$L_{entry} = (L_{exit}(s) \setminus kill(s)) \cup gen(s)$$

$$L_{exit}(1) = \{x\}$$

$$L_{exit}(2) = \{x\} \cup \{x\} = \{x\} \quad \text{sk 11}$$

$$L_{exit}(3) = \{x, y\}$$

$$L_{exit}(4) = \{x, y\} \cup \{x\} = \{x, y\} \quad \text{sk 6}$$

$$L_{exit}(5) = \{x\}$$

$$L_{exit}(6) = \{x, z\}$$

$$L_{exit}(7) = \{x, z\} \cup \{z\} = \{x, z\}$$

$$L_{exit}(8) = \{x, z\}$$

$$L_{exit}(9) = \{x\}$$

$$L_{exit}(10) = \emptyset$$

$$Wentby(1) = (\{x\} - \{x\}) \cup \phi = \phi$$

$$Wentby(2) = (\{x\} \setminus \phi) \cup \{x\} = \{x\}$$

$$Wentby(3) = (\{x, y\} \setminus \{y\}) \cup \{x\} = \{x\}$$

$$Wentby(4) = (\{x, y\} \setminus \phi) \cup \{y\} = \{x, y\}$$

$$Wentby(5) = (\{x\} \setminus \{x\}) \cup \{x, y\} = \{x, y\}$$

$$Wentby(6) = (\{x, z\} \setminus \{z\}) \cup \{x\} = \{x\}$$

$$Wentby(7) = (\{x, z\} \setminus \phi) \cup \{z\} = \{x, z\}$$

$$Wentby(8) = (\{x, z\} \setminus \{x\}) \cup \{x\} = \{x, z\}$$

$$Wentby(9) = (\{x\} \setminus \{z\}) \cup \{z\} = \{x, z\}$$

$$Wentby(10) = (\{\phi\} \setminus \phi) \cup \{x\} = \{x\}$$

Q2) Prove the following code using axiomatic semantics

$\{x = a \wedge y = b\}$

$t := x;$

$x := y;$

$y := t;$

$\{x = b \wedge y = a\}$

Precondition $\rightarrow \{x = a \wedge y = b\}$

assignment $\rightarrow t := x;$

$x := y;$

$y := t;$

postcondition $\rightarrow \{x = b \wedge y = a\}$

$y := t \Rightarrow t = a \wedge x = b$

$x := y \Rightarrow y = b \wedge t = a$

$t := x \Rightarrow y = b \wedge x = a$

$wp(x = a \wedge y = b) == \{x = a \wedge y = b\}$

Q.3] Apply symbolic execution on the following code.

$x := 3$

$y := x + 2$

if ($y > 4$) then

$z := y * 2$

else

$z := y + 3$

→

$x := 3$

$y := 3 + 2$

if ($y > 4$) then → this condition is always true executed if statement

$z := y * 2$

$z := 10$

∴ $x = 3$, $y = 5$, $z = 10$

Q4] Solve the following.

- What is fixed point theory in DFA?
- Explain with example how operational semantic evaluates the conditional branching?
- Prove using operational semantic rules:

$$x := 1$$

$$x := x + 2$$

- Prove the following code using checklist of axiomatic semantics.

{ $x = n$ }

while $x > 0$ do $x := x - 1$;

{ $x = 0$ }

→

① A fixed point of function is a value that does not change when function is applied.

- In DFA we compute information (live variable, reaching definitions etc) repeatedly until the result stop changing. This stable result is called fixed point.

eg. Loop goes on changing the values after final stable value not change. That is fixed point.

② Operational semantic explains how a program executes step by step. For conditional branching it shows how program chooses which branch to execute based on condition.

eg. if (B) then
 S1 ;
 else
 S2 ;

∴ B is the boolean expression. if B is true then S1 is executed. else S2 is executed if B is false.

- In operational Semantics, This is return as

• $\langle \text{Statement}, \text{state} \rangle \rightarrow \langle \text{new statement}, \text{state} \rangle$

ex-

$\langle S1, S \rangle \rightarrow S' \text{ if } B \text{ is true, } \langle S2, S \rangle \rightarrow S' \text{ if } B \text{ is false}$
 $\langle \text{if (B) } S1 \text{ else } S2, S \rangle \rightarrow S'$

③ Structural operational semantics :-

$$\frac{[[x \mapsto 1]]_{S_0} \rightarrow [[x \mapsto 1]]_{S_1}}{\langle x = 1, S_0 \rangle \rightarrow S_1}$$

$$\frac{[[x \mapsto 1]]_{S_1} \rightarrow [[x \mapsto 1+2+3]]_{S_1}}{\langle x = 1, x = x+2, S_0 \rangle \rightarrow \langle x = x+2, S_1 \rangle \rightarrow S'}$$

$$\frac{[[x \mapsto 1]]_{S_0} \rightarrow [[x \mapsto 1]]_{S_1} \quad \frac{[[x \mapsto 3]]_{S_1}}{\langle x \mapsto 1+2, S_1 \rangle \rightarrow S'}}{\langle x = 1, S_0 \rangle \rightarrow S_1}$$

$$\langle x = 1, x = x+2, S_0 \rangle \rightarrow S'$$

④ $\{P\} S \{C\}$

$P \Rightarrow x = n$

$C \Rightarrow x = 0$

$S \Rightarrow \text{while } (x > 0) \text{ do } x = x + 1$

invariance $\Rightarrow x \geq 0$

Invariant is true in starting
 $x \geq 0$

$(x = n)$, applying body rule $x = x + 1$;

$x = n + 1$;

$\therefore$ This never reaches to 0

The invariant remains true

But the loop goes infinite.

$\therefore$ The Precondition does not reach.